Science · GROW · Autumn 2 · Week 2 · GROW_SCI_A2_W02_P1

TA Brief

I can compare a disc and a sphere across viewpoints and explain spherical evidence.

At check-out, record E for comparing disc and sphere evidence against the pupil's target.

At check-out, record E for today's lesson against each pupil's target on the evidence app.

No cold calling: Invite a response, give thinking time, and accept a pass. Point to Reference A or B, rephrase, then offer a two-word choice.

During You do: Record actual **I · S · C** per task. Use the participation-loop prompt so the pupil responds to feedback with an adult, and mark **R** only after the pupil has responded.

Feedback codes VF · WS · I · NS + … · E · R · // · ?
Give one feedback code. My response is the pupil's improvement to the work.
Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Exit pupil voice
What helped you today, and what would you change?
Accept a pointed, spoken, signed, drawn, written or chosen response. A pupil may pass.

Stage timings

Stage	Minutes
Today I can	0
Arrival	5
Starter	4
I do	6
We do	7
You do	10
Review and improve	4
Exit ticket	4
Tools & print	0

Total: 40 minutes

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Guidance and staff card

Staff card · Science · GROW · Autumn 2 · Week 2 · GROW_SCI_A2_W02_P1

At check-out, record E for comparing disc and sphere evidence against the pupil's target.

At check-out, record E for today's lesson against each pupil's target on the evidence app.

Invite a response. Give thinking time. Every pupil may pass or share privately. Never use cold calling.

Return-to-learning route: Agree a quiet pause when needed, offer the regulation words beside the colour (Green · ready to start, Amber · need a small adjustment, Red · need a quiet pause), then re-enter through one low-demand visual step (pointing to one viewpoint row on Reference A) before resuming the route task.

Transitions and environment: Use soft instrumental music for transitions if the class plan says so. Keep the ball and rounded card disc on a stable tray.

Optional participation-loop prompt (inside You do, not a separate stage): The pupil responds to feedback with an adult, and R is marked only after the pupil has responded.

Feedback codes VF · WS · I · NS + … · E · R · // · ?
Give one feedback code. My response is the pupil’s improvement to the work.
Adult checks the response and ticks it, or writes R (Pupil responded to feedback).

Feedback and marking codes

Code	Meaning and staff use
VF	Verbal feedback given at this point
WS	Worked with adult support
I	Completed independently
NS + …	Next step: one specific, actionable improvement
E	Evidence captured on the evidence app
R	Responded: the pupil responds to feedback with an adult, and R is marked only after the pupil has responded
//	Pupil self-edit point, in the same colour on the same page
?	Read this back to me (adult checks spoken meaning with the pupil)

Keep the pupil's first attempt visible. Never require a special pen colour or box for corrections; edits happen in the same colour on the same page after **//**.

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P17 · SLT planning and evaluation

Lesson code: GROW_SCI_A2_W02_P1 · **Arc:** introduce-explore (Lesson 1 of 2: Introduce and explore)

Scheme reference: GROW Science Scheme of Work · Autumn 2 Week 2 · Describe the Sun, Earth and Moon as approximately spherical bodies (rebuilt from GROW W02 and GROW W9A with Key Stage 3 progression across 0°/45°/90° viewpoint geometry, controlled variables, spacecraft evidence, and model limitations).

Evidence in this lesson: Three-viewpoint comparison (0°, 45°, 90°) of a flat disc and a sphere, fairness controls, explanation of 'approximately spherical', and one model limitation in the book or usual mode, with I · S · C support mark. At check-out, record E for comparing disc and sphere evidence against the pupil's target. At check-out, record E for today's lesson against each pupil's target on the evidence app.

Planned adaptations

Area of need	Planned adaptation in this lesson	Where used
Cognition and learning	Three-viewpoint comparison table on Reference A (0°, 45°, 90°) paired with the space evidence table on Reference B	Arrival, I do, We do Comparator, Supported route
Communication and interaction	Point, sign, choice card, or adult exact-word scribing; optional participation-loop prompt with adult; adult read-back at Exit	Every stage check, You do, and Exit ticket
Social, emotional and mental health	Self-contained viewpoint diagrams on Reference A so no previous worksheet is required; regulation check with words beside colour	Arrival, I do, You do, Review
Sensory and physical	Static high-contrast SVG diagrams with ≥18 px labels; 44 px controls; large clean ball and stiff card disc with rounded edges on a tray	All slides and Reference A/B
Language access	Explicit sentence stems ('At the 90° viewpoint, the flat disc looks ___') and pre-taught Tier 2/Tier 3 vocabulary	I do word strip, Word help, Knowledge organiser

Where catch-up is embedded

1. **Arrival Q1** uses the self-contained Box 1 / Box 2 / Box 3 picture cue so any pupil can start immediately regardless of prior attendance. 2. **I do** models all three viewpoints (0°, 45°, 90°). 3. **Reference A and B** stay on the desk throughout You do. 4. The **Catch-up sheet** is used for absence or access only, replacing a starting step rather than adding extra workload.

If behind

Retain the Starter viewpoint check, the I do three-angle walkthrough, the We do Shape Viewpoint Comparator plus checkpoint question, one You do route task, and the Exit ticket. Use the supplied viewpoint diagrams on Reference A.

Staff evaluation (complete after teaching)

Evaluation prompt	Staff record
Delivered as planned	Yes / Partly / No — note any timing or route change
Success criteria met	Record which criteria were secure across Supported, Independent, Aspire, and Choice
Adaptations used and effect	Note which planned adaptations were used and how pupils responded
Coverage impact and next step	Ready for Autumn 2 Week 2 Lesson 2 (GROW_SCI_A2_W02_P2 — Show what you know)
Evidence banked	Book strip / work sample / At check-out, record E for comparing disc and sphere evidence against the pupil's target.

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P18 · Individual target opportunities

Select only the target that matches the pupil's existing individual plan held by the school. Do not copy personal details into this resource.

Target domain	Observable opportunity in this lesson	Adult recording note
Cognition and learning	Compares a flat disc and a sphere at 0°, 45°, and 90° and explains why one view is inconclusive	Record whether prompt was faded after the 0° view
Communication	Uses 'approximately spherical' and states one model limit in the pupil's agreed mode	Note mode used and whether adult read-back was confirmed
Regulation	Uses the regulation check or requests an adjustment and returns to one Reference A row	Follow agreed individual regulation plan
Sensory and physical	Uses the Shape Viewpoint Comparator or handles the ball and rounded card disc safely	Separate physical access support from science cueing
Independence	Completes the chosen route task with honest I, S, or C marking and optional participation loop	Mark actual independence per task; mark R only after pupil responds

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Adult deployment and responsive feedback

Two-adult deployment: During I do and We do, one adult leads the Shape Viewpoint Comparator while the second adult monitors access and private responses. During You do, one adult supports pupils on the Supported/Choice routes while the other circulates across Independent and Aspire, swapping roles across the week so no pupil becomes prompt-dependent on one adult.

Single-adult / cover deployment: Keep Reference A displayed on screen during You do. Set Independent and Aspire learners started on Task 1, then check in with Supported/Choice learners using the two-option stems before rotating back for VF and NS + … during You do and Review.

Responsive feedback and optional participation loop: Inside You do (or during Review), use the optional participation-loop prompt: give private VF and one specific NS + …. The pupil responds to feedback with an adult (making a **//** edit in the same colour on the same page in their book or explaining their reasoning orally), and **R** is marked only after the pupil has responded. At check-out, record E for comparing disc and sphere evidence against the pupil's target. At check-out, record E for today's lesson against each pupil's target on the evidence app.

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First lesson back · once-only read

1. Five-minute book scan before teaching

Scan books for five minutes to check pupils' Week 1 rotation work. Identify access barriers and actual starting points.

2. Existing NS + is history

Do not spend lesson time chasing historical feedback from previous weeks. Start fresh feedback from today's viewpoint comparison and // refinement.

3. Brief check with cover staff and TAs

Confirm that all tasks use the self-contained viewpoint diagrams on Reference A and the evidence summaries on Reference B so no pupil needs a previous worksheet.

4. Check the roll against evidence

If a pupil missed Week 1, give them Reference A and the Arrival picture cue; do not require them to complete missed sheets first.

5. One read after Exit, once

Read today's Exit ticket once against the three success criteria to adjust grouping for Lesson 2 (Show what you know).

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Teacher answers and topic guidance

Accept any scientifically equivalent wording in spoken, signed, pointed, drawn, or written form. Every question's wording is listed beside its number below, and the I do and We do model answers match the lesson page word for word.

Starter

1. Both a flat disc and a solid sphere look circular from the front (0°). Which test distinguishes them fairly?

Option A is right. Turn both objects to 45° and 90° at equal distance to compare their outlines.

I do model answer

1. How do we compare a disc and a sphere across viewpoints? Why do we describe Earth as approximately spherical?

Follow four steps in the Shape Evidence Lab. Step 1 (Front view at 0°): Both the flat disc and the sphere show a circular outline. One circular view cannot prove an object is a sphere. Step 2 (Oblique view at 45°): Keep viewing distance equal. At 45°, the disc looks oval while the sphere stays circular. Step 3 (Edge-on view at 90°): Turn both objects 90°. The disc shows a narrow edge while the sphere stays circular. Step 4 (Space evidence). Multi-viewpoint spacecraft photos show the Sun, Earth, and Moon are approximately spherical. Earth has mountains and a slight equatorial bulge. A classroom ball is a geometric model, not real planetary proof.

We do model answer and checkpoint

1. Inspect Cards A to D in the Shape Viewpoint Comparator. Match each card to its evidence role. Then repair the over-claimed statement.

Card A (Front view 0°: both look circular) shows why one view is inconclusive. Card B (45° and 90°: disc becomes oval then narrow) separates a disc from a sphere. Card C (Sphere in three views: circular at 0°, 45°, and 90°) models a 3D round body. Card D (Model limit and space evidence) explains why Earth is approximately spherical. Repair: do not claim one front photo proves a smooth sphere. Compare multiple viewpoints and state approximately spherical.

2. A flat disc and a solid ball turn 90° at equal distance. What outlines do we observe?

Option A: The flat disc looks narrow, while the solid ball keeps a circular outline.

Arrival · Supported

1. **1. Look at Box 1 on the left of the arrival drawing. Which object is a three-dimensional solid: the ball or the flat circle?** — The ball is a three-dimensional solid.
2. **2. Look at Box 1 again. If a mystery object looks circular from the front (0°), must it be a sphere?** — No. A flat disc also looks circular from the front.
3. **3. Look at Box 2 in the middle of the drawing. When the flat disc turns edge-on (90°), does its outline look narrow or wide?** — Its outline looks narrow when turned edge-on at 90°.
4. **4. Look at Box 3 on the right of the drawing. For a fair comparison, should both objects be held at the same distance?** — Yes. Keep both objects at the same viewing distance and angle.

Arrival · Independent

1. **1. Use Box 1 of the arrival drawing. Distinguish a three-dimensional sphere from a two-dimensional flat disc.** — A sphere has three-dimensional depth in all directions, while a flat disc is thin.
2. **2. Look at Box 1. Does one circular front view at 0° prove an object is a sphere?** — Both a flat disc and a solid sphere have a circular outline at 0°.
3. **3. Look at Box 2. How does a flat disc outline change at 45° and 90°?** — The disc becomes oval at 45° and narrow at 90°, whereas the sphere stays circular.
4. **4. Look at Box 3. Why do scientists call Earth 'approximately spherical' rather than a perfect smooth sphere?** — Earth has mountains and a small equatorial bulge, so it is slightly flattened.

Arrival · Aspire

1. **1. Use Box 1 of the arrival drawing. Explain geometrically why a flat disc and a sphere share the same outline at 0°.** — At 0°, the observer views the circular face perpendicular to the line of sight.
2. **2. Look at Box 2. Why does a 90° turn rule out a flat disc when an outline stays circular?** — A flat disc shows a narrow edge at 90°. A sphere stays circular from every angle.
3. **3. Look at Box 2 and Box 3. Name the two variables that must be controlled during a viewpoint comparison.** — Keep viewing distance and viewing angles (0°, 45°, 90°) identical for both objects.
4. **4. Look at Box 3. Distinguish what a classroom ball test shows from real spacecraft evidence of Earth.** — The ball models 3D geometry. Spacecraft images provide empirical evidence of the real Earth.

Arrival · Choice

1. **1. Look at Box 1 of the arrival drawing. Which object is three-dimensional: a ball or a flat card?** — A ball is three-dimensional.
2. **2. At 0° (front view), do both the disc and the ball look circular: Yes or No?** — Yes. Both look circular from the front.
3. **3. Look at Box 2. At 90° (edge-on), which object looks narrow: the flat disc or the ball?** — The flat disc looks narrow at 90°.
4. **4. Look at Box 3. Is Earth a perfect smooth ball or approximately spherical?** — Earth is approximately spherical.

You do · Supported mark scheme

1. **1. Use Reference A. Compare the flat disc and the solid sphere at 0° (front), 45° (oblique), and 90° (edge-on). Write or point to the outline for each view.** — At 0°, both look circular. At 45°, the disc looks oval and the sphere looks circular. At 90°, the disc looks narrow and the sphere looks circular. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Use Reference A. Complete: 'At 90°, the flat disc looks ___, while the sphere stays ___. We keep viewing distance the ___.'** — At 90°, the flat disc looks narrow, while the sphere stays circular. We keep viewing distance the same. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

3. **3. Use Reference B. Explain why we say Earth, the Moon, and the Sun are 'approximately spherical.' State one limit of the classroom ball.** — They are nearly round, but Earth has mountains and a slight bulge. Limit: a classroom ball is a model, not real space evidence. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

You do · Independent mark scheme

1. **1. Use Reference A. Record the outlines of a flat disc and a sphere across three controlled viewpoints (0°, 45°, and 90°). State the two fairness controls.** — At 0° both are circular; at 45° the disc is oval and the sphere is circular; at 90° the disc is narrow and the sphere is circular. Controls: same distance and same angles. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Use Reference A. Why does one circular front photo fail to prove a space body is a sphere?** — A flat disc and a sphere both produce a circular outline from the front (0°), so multiple viewpoints are required to distinguish them. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
3. **3. Use Reference B. Link multi-viewpoint space evidence and one model limit. Explain why Earth is called approximately spherical.** — Spacecraft observations from multiple viewpoints show Earth is round, and 'approximately spherical' accounts for mountains and the equatorial bulge. Limit: a desk ball omits real scale. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

You do · Aspire mark scheme

1. **1. Use Reference A. Contrast the outlines of a flat disc and a sphere at 0°, 45°, and 90°. Keep viewing distance equal.** — Because a disc lacks depth, rotating it from 0° to 45° and 90° foreshortens its circular face into an ellipse and then a narrow strip. A sphere's symmetry keeps a circular outline at every viewpoint. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. Use Reference B. Evaluate the claim: 'One circular image proves a body is a perfect sphere.' Separate observation from interpretation.** — Observation: one image shows a circular 2D outline. Interpretation error: a flat disc at 0° shares that outline, and real planets are approximately spherical (oblate) with surface relief. *(Maximum 3 marks. Accept equivalent valid wording or a clear model demonstration.)*
3. **3. Use Reference B. Link model evidence and spacecraft views to justify calling Sun, Earth, and Moon approximately spherical. State one model limit.** — The ball model predicts constant circular outlines across angles, matching multi-viewpoint spacecraft imagery of the Sun, Earth, and Moon. 'Approximately spherical' reflects Earth's equatorial bulge; the desk ball lacks real scale and gravity. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

You do · Choice mark scheme

1. **1. 1. Look at Reference A. At 0° (front view), do the disc and sphere look the same (circular) or different? 2. At 90° (edge-on), which object looks narrow: the disc or the sphere?** — 1. They both look circular at 0°. 2. The disc looks narrow at 90°. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*
2. **2. 3. Should viewing distance be the same or different for a fair test? 4. Can one front photo alone prove a sphere: No or Yes?** — 3. Viewing distance must be the same. 4. No,

because a disc also looks circular from the front. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

3. **3. 5. Look at Reference B. Which phrase best describes Earth, Sun, and Moon: approximately spherical or flat discs? 6. Name one limit of a smooth ball model: omits mountains/scale, or is real Earth?** — 5. Approximately spherical. 6. It omits mountains and real planetary scale. *(Maximum 2 marks. Accept equivalent valid wording or a clear model demonstration.)*

Exit ticket · Supported

1. **1. When turned edge-on at 90°, which object becomes narrow: a flat disc or a sphere?** — A flat disc becomes narrow at 90°, while a sphere stays circular.
2. **2. Why do we say Earth is 'approximately spherical' instead of a perfect smooth ball?** — Earth has surface mountains and is slightly flattened with a small bulge.
3. **3. Is a classroom ball a helpful model of Earth or a real photograph from space?** — It is a helpful classroom model, not a real space photograph.

Exit ticket · Independent

1. **1. Describe one observation across viewpoints (0°, 45°, 90°) that separates a flat disc from a sphere.** — When turned from 0° to 90°, a flat disc changes from circular to narrow, whereas a sphere keeps a circular outline.
2. **2. Why must viewing distance and angle stay equal when comparing the disc and sphere?** — Keeping distance and angle equal ensures a fair comparison so only the object's shape causes any outline change.
3. **3. Why are the Sun, Earth, and Moon called approximately spherical? State one model limit.** — Space observations show they are nearly round with slight flattening and surface features; a smooth desk ball omits scale and relief.

Exit ticket · Aspire

1. **1. Why are multi-viewpoint spacecraft photos stronger evidence than one circular photo?** — One circular view fits a flat disc at 0°, whereas circular outlines from multiple viewpoints confirm three-dimensional spherical geometry.
2. **2. Explain why describing Earth as 'approximately spherical' is more accurate than 'a perfect sphere'.** — Earth's rotation causes a small equatorial bulge (oblate spheroid) and its crust has mountains and valleys.
3. **3. State what the classroom disc-and-sphere comparison proves and what it cannot prove alone.** — It proves how 2D and 3D shapes project across angles, but real planetary shape requires empirical space observations.

Exit ticket · Choice

1. **1. At 90° (edge-on), does a sphere stay circular or turn into a thin line?** — A sphere stays circular at 90°.
2. **2. Does one front circle prove a sphere: No (a disc is circular too) or Yes?** — No. A flat disc also looks circular from the front.
3. **3. Are the Sun, Earth, and Moon flat discs or approximately spherical bodies?** — They are approximately spherical bodies.

Exit pupil voice

1. What helped you today, and what would you change? — Accept any pointed, spoken, signed, drawn, written, or chosen reflection on what helped and what could change next time. A pupil may pass.

Topic guidance and Key Stage 3 progression

- **Three-viewpoint geometry at Key Stage 3:** At **0° (front)** both a flat disc and a sphere show a circular outline. At **45° (oblique)** the disc foreshortens into an oval, and at **90° (edge-on quarter-turn)** the disc becomes narrow while the sphere stays circular.
- **Controlled comparison:** Both objects must be observed at the same distance and the same angles so only three-dimensional shape causes the outline difference.
- **Approximately spherical & model limits:** Multi-viewpoint spacecraft observations show the Sun, Earth, and Moon are **approximately spherical** (Earth is an oblate spheroid with mountains and an equatorial bulge; a classroom ball models 3D geometry but is not itself proof of Earth's shape).

Made by Matt

Teacher and cover plan

Lesson: Science · GROW · Autumn 2 · Week 2 · GROW_SCI_A2_W02_P1 · **Title:** Spherical bodies: introduce and explore · **Weekly arc:** Introduce and explore (Lesson 1 of 2)

Learning goal: I can compare a disc and a sphere across viewpoints and explain spherical evidence.

At check-out, record E for comparing disc and sphere evidence against the pupil's target.

At check-out, record E for today's lesson against each pupil's target on the evidence app.

Stage	Minutes	Pupil activity and teacher check
Today I can	0	Why is one circular front view not enough to prove that an object is a sphere?
Arrival	5	Use Question 1 to check picture access first. Accept pointing, speech, sign, or book responses.
Starter	4	Every pupil commits before the reveal. Check that pupils choose turning both objects to 45° and 90° at equal distance.
I do	6	Step through the four model stages using Show next step, linking 0°, 45°, and 90° outlines to spacecraft evidence and model limits.
We do	7	Invite each pupil to predict and explain before clicking Check card match, Check the repair, and Check checkpoint.
You do	10	One adult supports the agreed access group; the other circulates across routes. Mark I, S, or C beside each task to show actual independence.
Review and improve	4	Give private verbal feedback and one clear next step. Wait for the pupil response and engage with it before closing the loop and marking R.
Exit ticket	4	Receive each response and repeat the science back so the pupil can confirm or correct it.

Stage	Minutes	Pupil activity and teacher check
		Record E against the pupil's target at check-out.
Tools & print	0	Select only the sheet needed for access. Do not add catch-up work to a present pupil.

Total time: 40 minutes. **Next lesson:** Spherical bodies: show what you know.

Check-out and participation loop: Inside You do, use the optional participation-loop prompt (the pupil responds to feedback with an adult, and R is marked only after the pupil has responded). At check-out, record E for comparing disc and sphere evidence against the pupil's target.

Exit pupil voice
What helped you today, and what would you change?
Accept a pointed, spoken, signed, drawn, written or chosen response. A pupil may pass.

Equipment & safety: Reference A and B sheets, route cards, optional large clean ball and stiff card disc with rounded edges on a stable tray (do not use sharp skewers), exercise books, and pens/pencils (same pen colour used before and after //).

Made by Matt